REVISION LOG FOR "A CORRESPONDENCE FOR THE GENERALIZED HECKE ALGEBRA OF THE METAPLECTIC COVER $\overline{SL(2,F)}$, F p-ADIC"

DAVID JOYNER

1. Files and provenance

- Original: *New York J. Math.* **4** (1998), 223–235. Original source `howe6.tex`, with an AMS-LaTeX conversion `hecke-alg-corresp-for-metapletic-SL2-1998-amsart.tex` and the PDF `hecke-alg-corresp-for-metapletic-SL2_1998.pdf`.
- Revised paper: `hecke-alg-corresp-for-metapletic-SL2-revised.tex` (this log describes Draft 1 of it).
- Style guide: `joyner-math-paper-style-guide.md`, Draft 2, in the top-level folder of the repository.
- The file was edited in place, at the author's request.

2. AI usage

Three AI systems took part. None of them is listed as an author.

(1) **Argon (Google Gemini), reviser.** Argon made the 2026 mathematical corrections to the original text, before the present pass. The former `\thanks` note of the revised file summarized them as follows: corrections to the generator relations of Proposition 6.1 (formerly Proposition 8), explicit level and cocycle splitting conditions for general n in Proposition 7.2 (formerly Proposition 10), and a clarification of Savin's Iwahori–Hecke isomorphism in Section 9 (formerly Section 6). This log does not itemize Argon's changes line by line. A full comparison with the 1998 conversion is still to be done.

(2) **ChatGPT (OpenAI), referee.** ChatGPT refereed the corrected text. Its report is not in the repository.

(3) **Claude (Anthropic, Claude Opus 5.5), editor.** Claude made the Draft 1 changes listed in Sections 3 and 4 below. It brought the file into line with the style guide, and drafted new expository material. It made no change to any statement, proof, hypothesis or status of a result, except the rewording of one sentence recorded in Section 3.6. Claude did not referee the mathematics.

Review by AI systems is not an independent certification of the results.

3. Changes in Draft 1 (editorial)

3.1. Typesetting (guide Section 1).

- Class options `12pt,letterpaper,reqno` (class `amsart` kept). Added `fontenc` (T1), `newtxtext` and `newtxmath` (Times), `geometry` with letter paper and one-inch margins, `microtype`, and `array`/`longtable` for the notation table.

Date: Log entry 1, for Draft 1, November 11, 2026.

- Theorems, propositions, lemmas, corollaries, definitions, examples and remarks now share one counter numbered within sections. Remarks were previously on their own counter. Equations are numbered within sections.
- The hand-typed "As noted in Section 1" (old Section 4, now Section 7) became a `\ref` to Section 4.
- The final compile with `pdflatex` has no errors, warnings, undefined references or overfull boxes. The three overfull boxes of the previous version disappeared with the change to Times fonts and `microtype`. No manual line breaks were added.

3.2. **Title page, abstract and acknowledgments (guide Sections 5, 6.2, 6.3).**

- Subtitle added: "Draft 1. Corrected and expanded version". The original title is unchanged.
- Date line added: "Original draft: 1998. Corrected and expanded draft: 2026."
- The Naval Academy address was removed. The email address `wdjoyner@gmail.com` was kept.
- The `\thanks` note now identifies the original publication and the date of the revision.
- The abstract now opens with an AI-disclosure sentence and a pointer to the Acknowledgments. The rest of the abstract is unchanged.
- New unnumbered Acknowledgments section. The original thanks to Bill Wardlaw and the anonymous referee moved there word for word from the `\thanks` note. The AI credits, the non-certification statement and the pointer to this log were added.
- The sentence of the former `\thanks` note describing the 2026 corrections ("This revised version incorporates corrections to. . . ") was removed from the paper (guide 6.4). Its content is in Section 2 of this log.

3.3. **Citation keys (guide 1.7).** All 14 keys and bibliography labels were changed, and every `\cite` was updated (36 citations). The entries themselves are unchanged.

Old	New	Old	New
Bo	Bo76	KP	KP84
CR	CR81	M	Mo88
F	Fl80	RS	RS77
FK	FK86	S	Sa88
G	Ge76	Sch	Sc98
H	HM85	Sil	Si79
K	Ku67	W	Wa80

Nine references were added for the literature review: BK98, CL25, GGW18, GS12, Ka21, Pe13, Sa04, TW18 and We18.

3.4. **Structure (guide Section 2).**

- The unsectioned opening text became Section 1, Introduction, with an orienting paragraph, the running example and a paragraph on the organization of the paper.
- The paragraph of the opening text that surveys earlier work ("The study of genuine admissible representations. . . ") moved, word for word, to the new Section 2.1. Section 2 is a new literature review, split into work up to 1998 and work after 1998.
- New Section 3, Background, with four subsections.
- The old Section 1, "Some notation", is now Section 4. A table of notation was added at its start.
- Old Sections 2–6 are now Sections 5–9.

- New Section 10, Summary of results, and Section 11, Glossary, followed by the Acknowledgments and the references.

3.5. **Orienting paragraphs (guide 3.2).** New orienting paragraphs were added at the start of Sections 1, 2, 2.2, 3, each subsection of 3, and Sections 4, 5, 8, 9, 10 and 11. Sections 6 and 7 already began with an orienting paragraph, and Section 2.1 begins with one drawn from new text.

3.6. **Wording fix.** In Section 4 the sentence "In fact, directly from the definition of β, for any $g_1, g_2 \in B_k$, the Iwahori factorization $B_k = U_k^- T_k U_k^+$ implies that $\overline{B}_k$ splits." was not grammatical. At the author's request it now reads "In fact, the definition of β and the Iwahori factorization $B_k = U_k^- T_k U_k^+$ imply directly that $\overline{B}_k$ splits." The dangling phrase "for any $g_1, g_2 \in B_k$" was dropped. The claim is unchanged.

4. New mathematical and expository text, for refereeing (ground rule 2)

Every item below is new in Draft 1 and was written by Claude. None of it changes or adds to the results of the paper. Each item should be refereed.

(1) **Example 1.1** (running example: $F = \mathbb{Q}_3$, $n = 2$, $k = 1$). Values of the quadratic Hilbert symbol, including $(-1, 3) = -1$.
(2) **Section 2** (literature review). Summaries of BK98, Sa04, GS12, TW18, CL25, Ka21, Pe13, We18 and GGW18 are based on their abstracts only. The claim that the dual group in Weissman's framework depends on the parity of n, in agreement with Section 9, is from memory (verification level 3).
(3) **Section 3.1** and **Example 3.1** ($\mathcal{H}(F^\times//O^\times) \cong \mathbb{C}[t, t^{-1}]$).
(4) **Section 3.2** and **Example 3.2**: properties of the Hilbert symbol, $\beta(h(a), h(b)) = (b, a)$, $\beta(h(-1), h(3)) = -1$ and $\beta(s_1, s_1) = (-1, -1) = 1$ in $\mathbb{Q}_3$.
(5) **Section 3.3** and **Example 3.3**: $[B : B_1] = q^2(q-1) = 18$, $\mathrm{vol}(B_1 s_1 B_1) = 3$, $\mathrm{vol}(B_1 s_1 s_2 B_1) = 9$.
(6) **Section 3.4** and **Example 3.4**: the description of Borel's produced module as the smooth part of $\mathrm{Hom}_{\mathcal{H}(H//K)}(C_c(K\backslash H), W)$ is from memory and should be checked against Bo76. The trivial representation and its module.
(7) **Notation table** at the start of Section 4.
(8) **Examples 4.2, 4.3, 4.4, 4.6** (for Definition 4.1, trivial cocycles, genuine functions, Definition 4.5).
(9) **Examples 5.2 and 5.4** (for Lemmas 5.1 and 5.3). Example 5.4 uses Lemma 6.3(d) to show $f_{s_1} * f_{s_1} \neq f_{-I}$.
(10) **Example 6.4** (relation A(i) in the running example, checked against the trivial representation).
(11) **Example 7.3** (Proposition 7.2 in the running example).
(12) **Examples 8.3, 8.5, 8.8, 8.11, 8.13** (main theorem, matrix coefficients, induced modules, supercuspidal and square-integrable representations, Corollary 8.12). Example 8.11 uses two standard facts stated without citation: the Steinberg representation of $SL(2, F)$ is square-integrable, and a representation with a nonzero Iwahori-fixed vector is a subquotient of an unramified principal series and so is not supercuspidal.
(13) **Example 9.3**: the Iwahori–Matsumoto quadratic relation $T_s * T_s = (q-1)T_s + qT_1$ for $PGL(2, \mathbb{Q}_3)$, stated without citation (verification level 3).
(14) **Section 10** (summary of results and open questions) and **Section 11** (glossary).

5. Concordance of numbering (guide 6.8)

All \labels were kept. The old numbers are those of the previous compiled version of the revised file, which agree with the 1998 paper.

Label	Old	New
(sections)	Section 1 Some notation	Section 4
	Section 2 Some lemmas	Section 5
section:sl2	Section 3	Section 6
section:metaplectic	Section 4	Section 7
	Section 5 Application	Section 8
section:savin	Section 6	Section 9
def:1	Definition 1	Definition 4.1
def:2	Definition 2	Definition 4.5
lemma:3	Lemma 3	Lemma 5.1
lemma:general	Lemma 4	Lemma 5.3
lemma:genuine	Lemma 5	Lemma 5.5
lemma:convolve	Lemma 6	Lemma 5.6
lemma:isomorphic	Lemma 7	Lemma 5.7
prop:main	Proposition 8	Proposition 6.1
rem:1	Remark 1	Remark 6.2
lemma:9	Lemma 9	Lemma 6.3
eq:obvious	equation (1)	equation (6.1)
rem:torus_center	Remark 2	Remark 7.1
prop:main2	Proposition 10	Proposition 7.2
thrm:main	Theorem 11	Theorem 8.1
corollary:main	Corollary 12	Corollary 8.2
def:13	Definition 13	Definition 8.4
prop:silberger	Proposition 14	Proposition 8.6
def:15	Definition 15	Definition 8.7
prop:borel	Proposition 16	Proposition 8.9
def:17	Definition 17	Definition 8.10
cor:18	Corollary 18	Corollary 8.12
thm:savin	Theorem 19	Theorem 9.1
cor:20	Corollary 20	Corollary 9.2

6. Original and new results (guide 6.5)

Every numbered theorem, proposition, lemma, corollary, definition and remark is from the original 1998 paper, as corrected by Argon in 2026 (Section 2). Draft 1 establishes no new results. The examples, the background, the literature review, the summary and the glossary are new exposition (Section 4).

7. Sources not consulted (guide 4.3)

Claude consulted none of the 14 original references, so no citation in the paper was checked against its source. In particular, these precise citations are unchecked: Ku67 §1 and Theorem 2,

HM85 Chapter 3 (Proposition 2.2, Theorem 2.1, page 33 and pages 38–39), Mo88 Proposition 3.4, Si79 Corollary 1.10.6, CR81 Lemma 3.14, and Bo76 §§1–2.

For the nine new references, the bibliographic data (authors, title, journal, volume, year and pages) were checked against the publisher's or arXiv's web page, except BK98 and Sa04, which were taken from the reference list of GS12 as shown on the AMS site. The summaries rely on the abstracts only. The full texts were not read.

8. Points for the author and the referees

These were noticed during the editorial pass. None was changed, because each would need a mathematical change (ground rule 1).

(1) Lemma 5.7 assumes only $s(1) = 1$. The paragraph before it also requires $s|_{B_k} \equiv 1$, which seems to be needed for ϕ to preserve bi-B_k^*-invariance.
(2) Proofs that are omitted or only sketched (guide 3.7): C(i) in the proof of Proposition 7.2 ("is omitted"), C(ii) and C(iii) there (sketched), the proofs of Propositions 8.6 and 8.9 ("one need only check"), the second part of Corollary 8.12 ("omitted"), and Corollary 9.2, which has no proof.
(3) Remark 6.2(3) ("Possibly B(iii) is a superfluous relation") is a speculation. It is listed as an open question in Section 10.
(4) The symbols $U_k^{\pm}$ and T_k are used in Section 4 before Lemma 6.3 and Remark 7.1 define them (guide 3.5). The new notation table now lists them before their first use. The text itself was not changed.

9. Style-guide rules: defaults and exceptions

[DECIDE] defaults used. Document class: kept `amsart`. Times package: `newtx`. Glossary placement: notation near the beginning (Section 4), glossary at the end just before the acknowledgments (Section 11). AI systems on the author line: not listed, with the credit in the acknowledgments.

Rules not applied, and why.

- The paper-specific rules in guide Section 9 for the metaplectic Jacquet–Langlands series (status labels (P), (E), (C) and so on, hypothesis tags, the $|\cdot|_p$ convention, a new file for each draft). This is a standalone 1998 paper stored in that repository. It is not part of the series. At the author's request, the file was edited in place.
- Rule 6.6 ("Remark 2026.n"). Choosing what to mark would require a verification that this editorial pass did not do. The candidates are listed in Section 8, item 2.
- Rules 3.6 and 3.7 (state all hypotheses, do not skip steps). Meeting them would change statements or proofs. See Section 8.
- Rule 3.3 (an example in every section) was not applied to the literature review, the summary of results or the glossary, which explain no new concepts.
- Rule 3.8 (no semicolons) was applied to new text only. Existing semicolons were kept.
- Rule 8.6 (copyright). The repository is public and already contains the corrected version of the published paper. The copyright position has not been checked.

Email address: wdjoyner@gmail.com